

01VIXSM

Deep Natural Language Processing

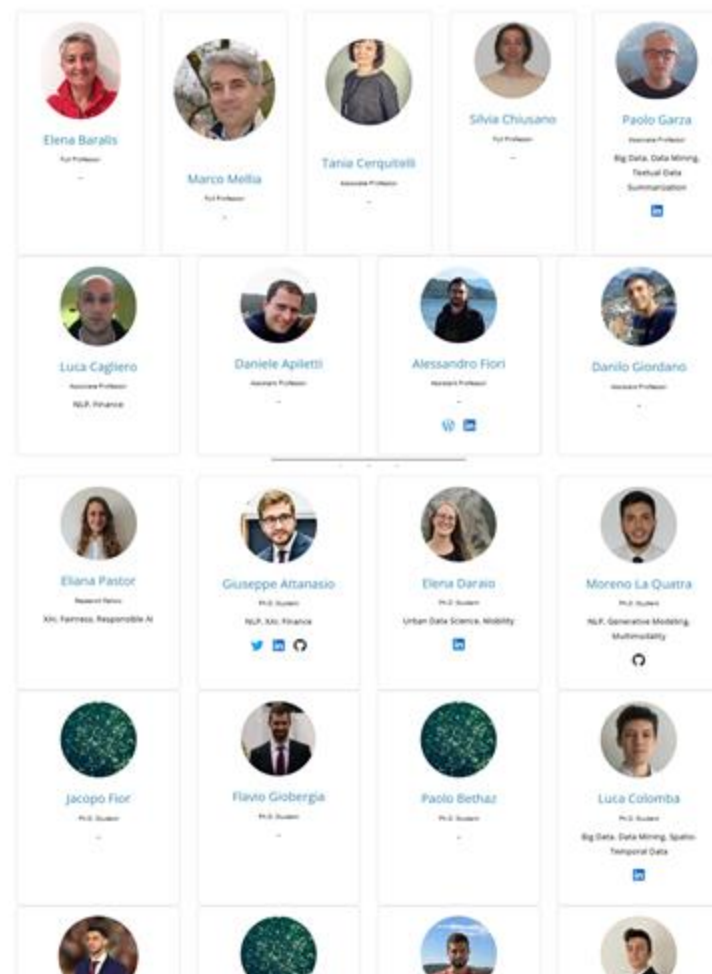
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**Politecnico
di Torino**

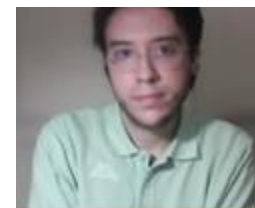
Who we are

- DBDMG group (<https://dbdmg.polito.it>)
- 4 Full Professors
- 4 Associate Professors
- 1 Researchers with Tenure Track
- 3 Researchers with Non-Tenure Track
- Around 20 PhD/Post Doc



NLP, Multimodal Learning and related areas

- Luca Cagliero, Associate Professor
- Lorenzo Vaiani, Post-Doc
 - Multimodal Learning
- Aurora Gensale, PhD Student
 - Agentic AI, Cybersecurity
 - Apprenticeship PhD, DriveSec
- Giuseppe Gallipoli, PhD Student
 - NLP
- Davide Napolitano, PhD Student
 - Multimodal Learning, Explainable AI
- Simone Papicchio, PhD Student
 - Tabular Representation Learning
 - Joint program with EURECOM (Sophia Antipolis, France)
 - Co-supervised by Prof. Paolo Papotti
- Yassine Ali, PhD Student
 - Time series Analysis
 - Tierra Spa



Current research interests

- Time-aware Large Language Models -> new project
- Text, Video, and Speech Summarization
- Document Question Answering
- Text Style Transfer
- Entity Recognition and Disambiguation from financial documents
- Fake news and hate speech detection
- Neural Shapley-based Explanation
- Anomaly Detection from Time Series
- Agentic AI frameworks

Calendar

- 8 CFU -> 80 hours of frontal lecture/practice
 - 41 hours of frontal lecture
 - 18 hours of in-class practice
 - In presence, except for
 - Catch-up lessons
 - We will use BBB Virtual Classroom if necessary
 - 21 hours of laboratory practices (mostly 3 hours each)
 - All in presence

Calendar

- Planning

- 14 weeks
- 7 labs -> half of the weeks
- 59 hours of in-class lectures/practices
 - Ideally
 - 4.5 hours per week
 - In practice
 - 6 hours per week for the first weeks of the semester
 - 3 hours in the last weeks
- Why?
 - We need to introduce NLP fundamentals asap to start the project work earlier
 - We will cancel some scheduled lectures in October

Original timetable (without changes/extra)

Orario Ricerca Avanzata

Stampa

	lunedì 22/09/2025	martedì 23/09/2025	mercoledì 24/09/2025	giovedì 25/09/2025	venerdì 26/09/2025
8 ⁰⁰					
9 ⁰⁰					
10 ⁰⁰					
11 ⁰⁰					
12 ⁰⁰					Deep natural language proce... CAGLIERO LUCA AA - ZZ LADISPE
13 ⁰⁰		Deep natural language proce... CAGLIERO LUCA AA - ZZ 3I			
14 ⁰⁰			Deep natural language proce... CAGLIERO LUCA AA - ZZ 3I Lezione/Esercitazione		
15 ⁰⁰					
16 ⁰⁰					
17 ⁰⁰					

Week#1 timetable variations

- Week#1 lectures
 - Tuesday 23/9 from 1pm to 2.30pm Room 3I
 - Wednesday 24/9 from 3pm to 5.30pm Room 3I
 - Thursday 25/9 from 8:30am to 10am online (BBB)
- Practices
 - NO LAB IN THE FIRST WEEK!

Week#2 timetable variations

- Lectures

- Tuesday 30/9 from 1pm to 2.30pm Room 3I
- Thursday 30/9 from 4pm to 7pm online (BBB)
- Wednesday 1/10 from 2.30pm to 5.30pm Room 3I

- Practices

- Friday 3/10 from 11.30am to 2.30pm LADISPE
 - I am trying to book a large room/lab (I'll keep you posted)

Videorecordings

- In-class lectures will be recorded using the Virtual Classroom (BBB)
- Video recordings will be made available to enrolled students only
 - The publication delay is unpredictable (typically, one or two days)
- Lab practices will not be recorded
 - We will share all the necessary materials (text, solutions, etc) through the Didactic Portal in advance

Videorecordings

- Limited interactions with remote students
 - We foster students' participation to in-presence lectures
 - Dual-channel communication (both remote and in-presence) degrades the quality of in-presence lectures
 - If you have specific questions, please send an email
- Students' microphones and cameras will be disabled

Prerequisites

- Fundamentals of Data Sciences, Machine Learning, and Deep Learning
- Basic knowledge of the Python language

Content outline

- Fundamentals of Natural Language Processing
- Vector representations of text
- Transformers
- Fundamentals of recommender systems
- Large Language Models
- LLM-based Agents
- Main NLP applications
 - Summarization
 - Machine Translation
 - AI Chatbots
 - ...

Main topics

- More details available at the official course webpage

https://didattica.polito.it/pls/portal30/gap.pkg_guide.viewGap?p_cod_ins=01VIXSM&p_a_acc=2026&p_header=S&p_lang=IT&multi=N

Laboratory topics

1. Text preprocessing and topic modelling
2. Word and sentence embeddings
3. Entity Recognition and intent detection
4. Information Retrieval and recommender systems
5. Summarization
6. Machine translation
7. AI Chatbots
8. LLM-based Agents

LABS START IN THE 2ND WEEK

(schedule of the next labs to be announced)

Laboratories

- We use Google Colab
 - <https://colab.research.google.com/>
- You need a Google account
- Practice text and solutions will be shared through the didactic portal to all enrolled students

Teaching materials

- Handouts will be shared through the didactic portal
- Additional materials
 - Scientific papers used to deepen the knowledge on specific topics covered by the slides
 - Not mandatory to pass the exam

Useful books

- Dan Jurafsky and James H. Martin. Speech and Language Processing (3rd ed. draft)
- Jacob Eisenstein. Natural Language Processing
- Yoav Goldberg. A Primer on Neural Network Models for Natural Language Processing
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning

Exam rules

- The exam consists of
 - a written test (max. 22 points)
 - Covering theoretical aspects introduced during the course
 - Mix of closed and open-ended questions
 - A group project (max. 10 points)
 - A team project is assigned during the course
 - The project must be submitted through the Didactic Portal (section "Elaborati")
 - The evaluation comprises an oral discussion
 - The final score is given by the sum of the points achieved in the written part and in the evaluation of the final report

Exam rules

- The written test is scheduled during the official exam session
 - All marks will be recorded by the end of the session
 - Exam booking is mandatory
 - The exam is closed-books
- The group project will be assigned well before the end of the course
 - Teams consist of 4-5 students
 - We collect team preferences regarding the candidate topics
 - The ultimate assignment of the specific topic is at the discretion of the teachers

Exam rules

- The project score is valid until the end of the academic year (until September 2026)
- When you pass the written exam we automatically record the final grade by summing the achieved project score (if any)
- If you want to improve the mark of the written test you can reject it and redo the test at the next available exam
- If you want to improve the score of the project you have to redo it from scratch
 - No supplementary oral part is allowed

Project rules

- Each team has to
 - Submit a project report (including references to project code, data, and models for reproducibility reasons)
 - Orally present the work (in group)
 - Whether using presentation slides or not
- To sum up the achieved project score within a exam session, you have to deliver the project report and code 1 week before the end of the examination session
 - We will revise report and code and set up an appointment for the discussion by the end of the session
- All grades of the written part (possibly including the project score) **MUST** be recorded within the official exam session

Project rules: example nr. 1

- I plan to give the written exam in the winter examination session (January-February 2026)

EXAM SESSION A.Y. 2025/2026

Date	1 ↑	* Event	2 ↑
Monday, 12/01/2026 to Saturday, 21/02/2026		WINTER EXAMINATION SESSION – a.y. 2025/26	
Monday, 08/06/2026 to Saturday, 18/07/2026		SUMMER EXAMINATION SESSION - a.y. 2025/26	
Monday, 31/08/2026 to Saturday, 19/09/2026		AUTUMN EXAMINATION SESSION – a.y. 2025/26	

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To get the extra points my team has to deliver the project at least one week before the end of the examination session, i.e., by 14 Feb 2026

Project rules: example nr. 2

- I plan to give the written exam in the summer examination session (June-July 2026)

Summer examination session
(June - July)
08 Jun 2026 - 18 Jul 2026

To get the extra points my team has to deliver the project at least one week before the end of the examination session, i.e., by 11 July 2026

Consulting

- At the end of the lecture or by setting up a dedicated appointment
 - Please send an email
- Practical questions about code and libraries
 - Mainly during the laboratory activities and in-class practices
- Theoretical/methodological questions
 - Mainly during frontal lectures

Consulting

- 2/3 in-class lectures will be exclusively devoted to consulting for the final project
 - Mainly for methodological questions and answers
 - No real-time code debugging!
- We recommend students to start the project work as soon as possible to take advantage of the consulting sessions

Acknowledgements and copyright license



- Copyright licence
 - Attribution + Noncommercial + NoDerivatives
- Affiliation
 - The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
 - <https://dbdmg.polito.it>
 - <https://smartdata.polito.it>

Thank you!